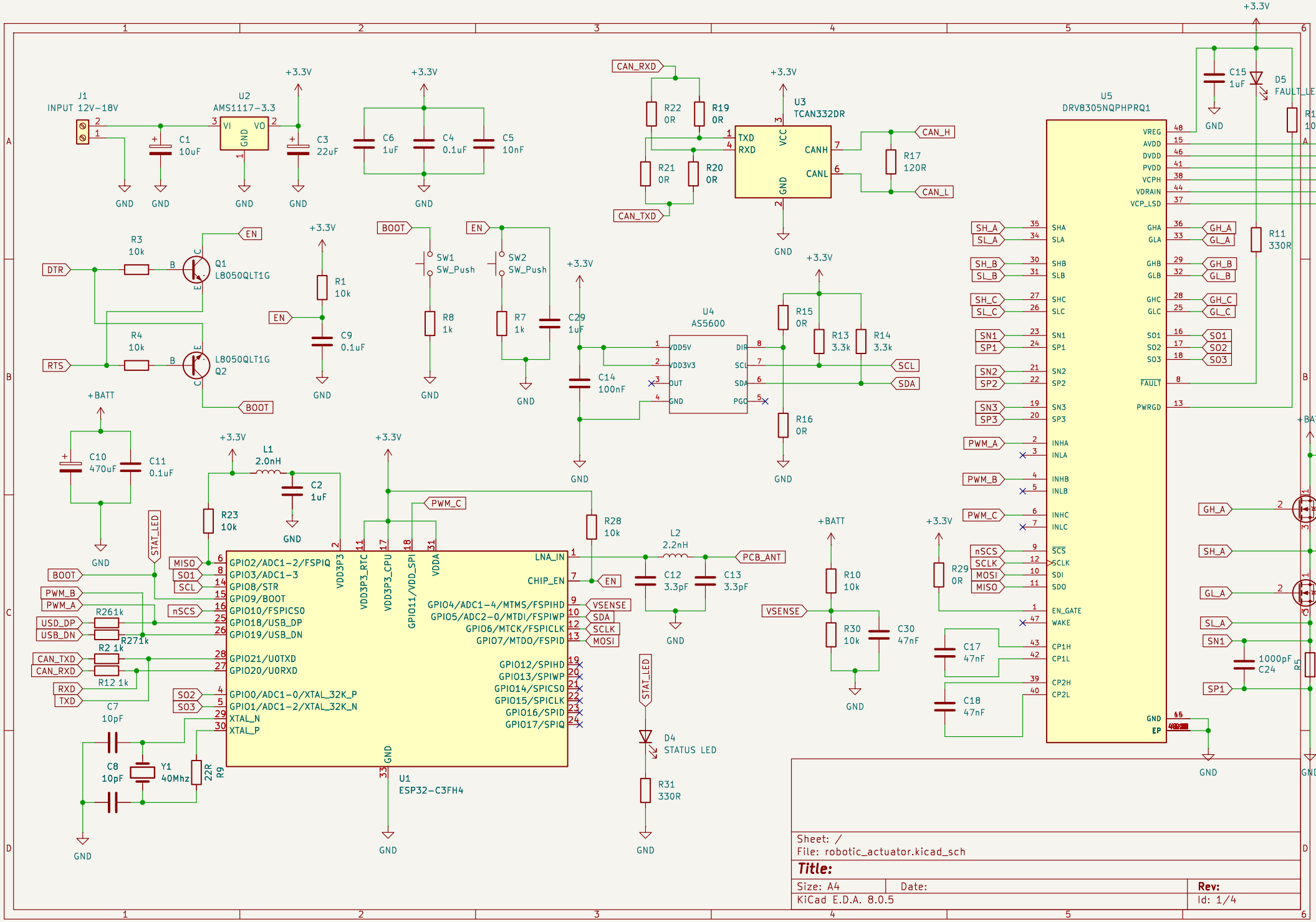
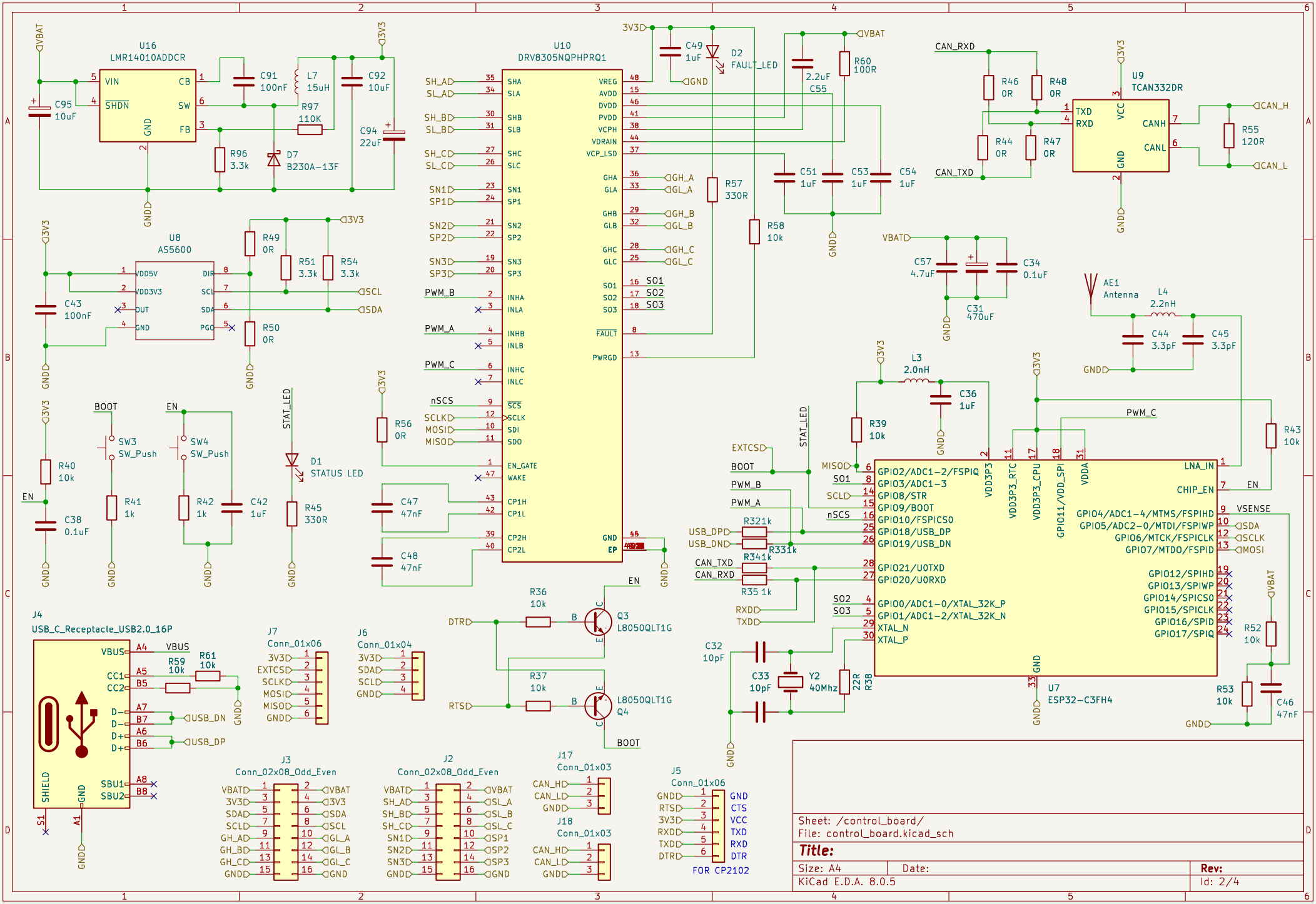
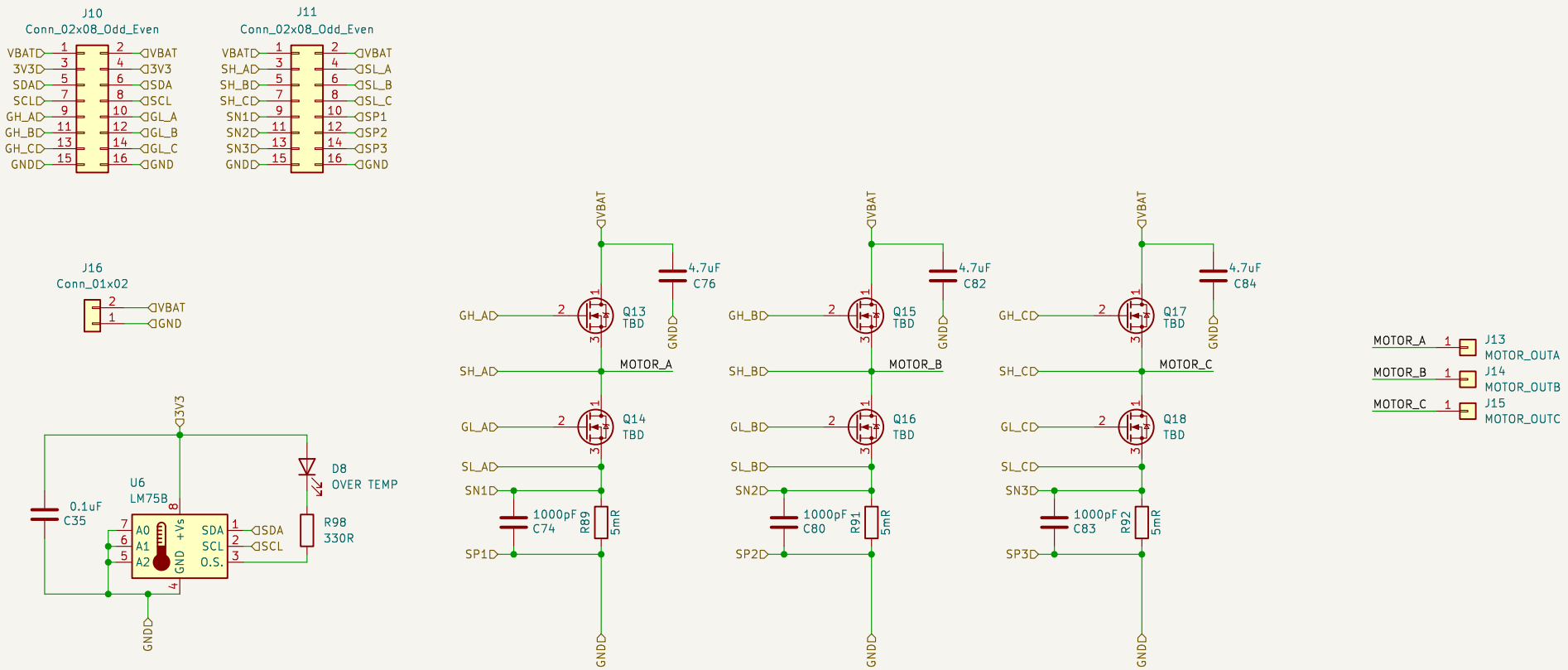

J1
INPUT 12V–18V
U2
AMS1117–3.3
C1
10uF
C3
22uF
C6
1uF
C4
0.1uF
C5
10nF
U3
TCAN332DR
R22
0R
R19
0R
R21
0R
R20
0R
R17
120R
CAN_RXD
CAN_TXD
CAN_H
CAN_L
U5
DRV8305NQPHPRQ1
C15
1uF
D5
FAULT_LED
R18
10k
R11
330R
R3
10k
R4
10k
Q1
L8050QLT1G
Q2
L8050QLT1G
DTR
RTS
EN
BOOT
R1
10k
C9
0.1uF
SW1
SW_Push
SW2
SW_Push
R8
1k
R7
1k
C29
1uF
C14
100nF
U4
AS5600
R15
0R
R13
3.3k
R14
3.3k
R16
0R
SCL
SDA
C10
470uF
C11
0.1uF
L1
2.0nH
C2
1uF
R23
10k
STAT_LED
PWM_C
R28
10k
L2
2.2nH
C12
3.3pF
C13
3.3pF
PCB_ANT
R10
10k
R30
10k
C30
47nF
VSENSE
R29
0R
C17
47nF
C18
47nF
Q5
TBD
Q6
TBD
GH_A
SH_A
GL_A
SL_A
SN1
SP1
C24
1000pF
R5
5mR
R261k
R271k
R2 1k
R12 1k
C7
10pF
C8
10pF
Y1
40Mhz
R9
22R
D4
STATUS LED
R31
330R
U1
ESP32–C3FH4
GPIO2/ADC1–2/FSPIQ
GPIO3/ADC1–3
GPIO8/STR
GPIO9/BOOT
GPIO10/FSPICS0
GPIO18/USB_DP
GPIO19/USB_DN
GPIO21/U0TXD
GPIO20/U0RXD
GPIO0/ADC1–0/XTAL_32K_P
GPIO1/ADC1–2/XTAL_32K_N
XTAL_N
XTAL_P
LNA_IN
CHIP_EN
GPIO4/ADC1–4/MTMS/FSPIHD
GPIO5/ADC2–0/MTDI/FSPIWP
GPIO6/MTCK/FSPICLK
GPIO7/MTDO/FSPID
GPIO12/SPIHD
GPIO13/SPIWP
GPIO14/SPICS0
GPIO15/SPICLK
GPIO16/SPID
GPIO17/SPIQ
VDD3P3
VDD3P3_RTC
VDD3P3_CPU
GPIO11/VDD_SPI
VDDA
MISO
SO1
SCL
nSCS
USD_DP
USD_DN
CAN_TXD
CAN_RXD
RXD
TXD
SO2
SO3
VSENSE
SDA
SCLK
MOSI
+3.3V
+BATT
GND
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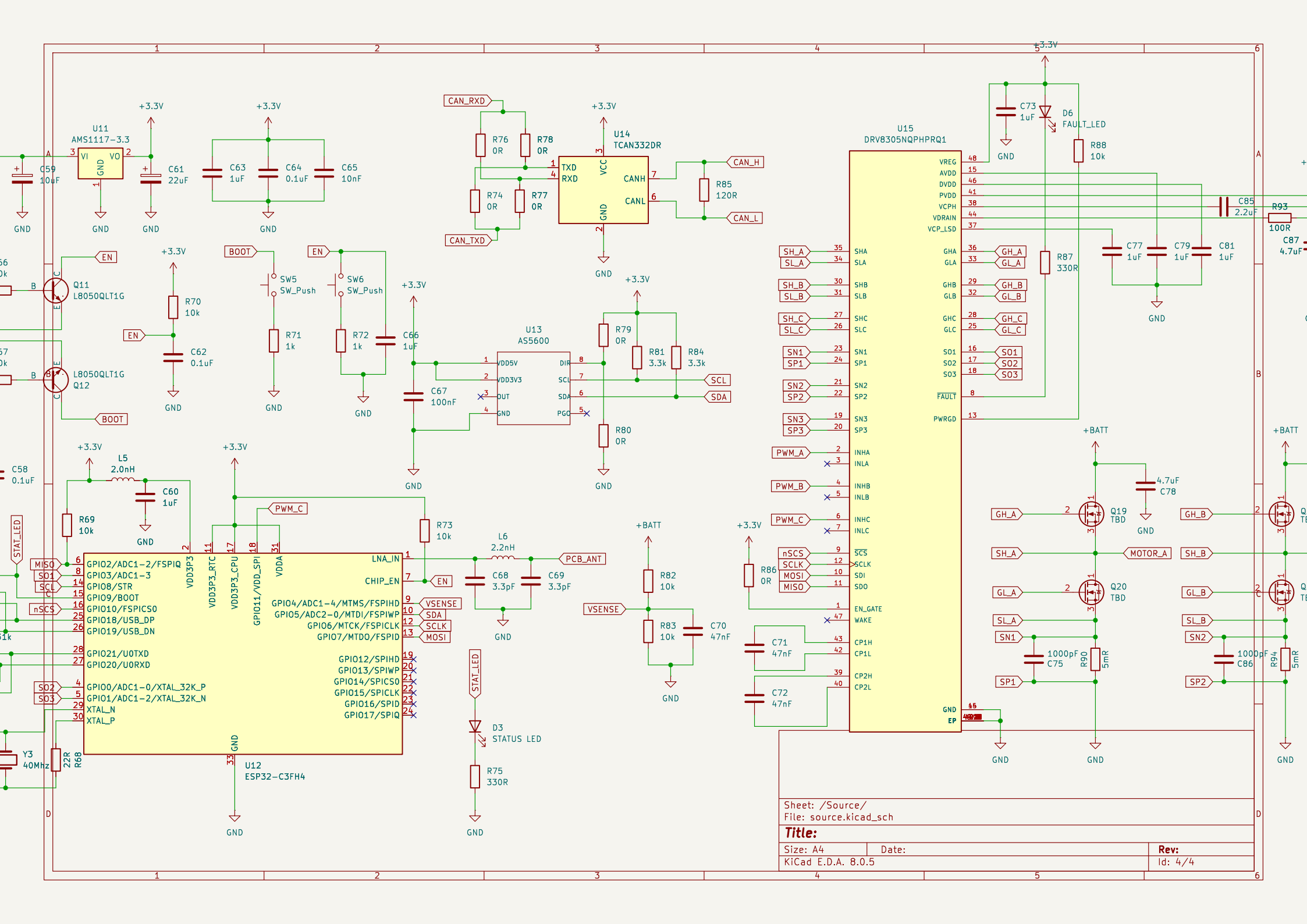

U16
LMR14010ADDCR
VIN
SHDN
CB
SW
FB
GND
C95 10uF
C91 100nF
L7 15uH
C92 10uF
C94 22uF
R97 110K
R96 3.3k
D7 B230A-13F
U8
AS5600
VDD5V
VDD3V3
OUT
GND
DIR
SCL
SDA
PGO
C43 100nF
R49 0R
R51 3.3k
R54 3.3k
R50 0R
SCL
SDA
U10
DRV8305NQPHPRQ1
SHA
SLA
SHB
SLB
SHC
SLC
SN1
SP1
SN2
SP2
SN3
SP3
INHA
INLA
INHB
INLB
INHC
INLC
SCS
SCLK
SDI
SDO
EN_GATE
WAKE
CP1H
CP1L
CP2H
CP2L
VREG
AVDD
DVDD
PVDD
VCPH
VDRAIN
VCP_LSD
GHA
GLA
GHB
GLB
GHC
GLC
SO1
SO2
SO3
FAULT
PWRGD
GND
EP
SH_A
SL_A
SH_B
SL_B
SH_C
SL_C
PWM_B
PWM_A
PWM_C
nSCS
SCLK
MOSI
MISO
C49 1uF
D2 FAULT_LED
2.2uF C55
R60 100R
R57 330R
R58 10k
C51 1uF
C53 1uF
C54 1uF
R56 0R
C47 47nF
C48 47nF
U9
TCAN332DR
TXD
RXD
VCC
CANH
CANL
GND
CAN_RXD
CAN_TXD
R46 0R
R48 0R
R44 0R
R47 0R
R55 120R
CAN_H
CAN_L
VBAT
C57 4.7uF
C31 470uF
C34 0.1uF
AE1
Antenna
L4 2.2nH
C44 3.3pF
C45 3.3pF
L3 2.0nH
C36 1uF
R39 10k
R43 10k
PWM_C
STAT_LED
EXTCS
BOOT
PWM_B
PWM_A
nSCS
USB_DP
USB_DN
R321k
R331k
R341k
CAN_TXD
CAN_RXD
R35 1k
RXD
TXD
SO1
SO2
SO3
C32 10pF
C33 10pF
Y2 40Mhz
22R R38
U7
ESP32-C3FH4
GPIO2/ADC1-2/FSPIQ
GPIO3/ADC1-3
GPIO8/STR
GPIO9/BOOT
GPIO10/FSPICS0
GPIO18/USB_DP
GPIO19/USB_DN
GPIO21/U0TXD
GPIO20/U0RXD
GPIO0/ADC1-0/XTAL_32K_P
GPIO1/ADC1-2/XTAL_32K_N
XTAL_N
XTAL_P
VDD3P3
VDD3P3_RTC
VDD3P3_CPU
GPIO11/VDD_SPI
VDDA
LNA_IN
CHIP_EN
GPIO4/ADC1-4/MTMS/FSPIHD
GPIO5/ADC2-0/MTDI/FSPIWP
GPIO6/MTCK/FSPICLK
GPIO7/MTDO/FSPID
GPIO12/SPIHD
GPIO13/SPIWP
GPIO14/SPICS0
GPIO15/SPICLK
GPIO16/SPID
GPIO17/SPIQ
VSENSE
SDA
SCLK
MOSI
R52 10k
C46 47nF
R53 10k
R40 10k
R41 1k
R42 1k
C42 1uF
C38 0.1uF
R45 330R
SW3 SW_Push
SW4 SW_Push
BOOT
EN
D1 STATUS LED
R36 10k
R37 10k
Q3 L8050QLT1G
Q4 L8050QLT1G
DTR
RTS
J4
USB_C_Receptacle_USB2.0_16P
VBUS
CC1
CC2
D-
D+
SHIELD
GND
SBU1
SBU2
R59 10k
R61 10k
J7
Conn_01x06
J6
Conn_01x04
J3
Conn_02x08_Odd_Even
J2
Conn_02x08_Odd_Even
J17
Conn_01x03
J18
Conn_01x03
J5
Conn_01x06
GND
CTS
VCC
TXD
RXD
DTR
FOR CP2102
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U11
AMS1117-3.3
C59 10uF
C61 22uF
C63 1uF
C64 0.1uF
C65 10nF
U14
TCAN332DR
R76 0R
R78 0R
R74 0R
R77 0R
R85 120R
CAN_RXD
CAN_TXD
CAN_H
CAN_L
U15
DRV8305NQPHPRQ1
C73 1uF
D6
FAULT_LED
R88 10k
R87 330R
C77 1uF
C79 1uF
C81 1uF
C85 2.2uF
R93 100R
C87 4.7uF
Q11
L8050QLT1G
Q12
L8050QLT1G
R70 10k
C62 0.1uF
SW5
SW_Push
R71 1k
SW6
SW_Push
R72 1k
C66 1uF
C67 100nF
U13
AS5600
R79 0R
R81 3.3k
R84 3.3k
R80 0R
SCL
SDA
EN
BOOT
L5
2.0nH
C60 1uF
C58 0.1uF
R69 10k
R73 10k
PWM_C
L6
2.2nH
C68 3.3pF
C69 3.3pF
PCB_ANT
R82 10k
R83 10k
C70 47nF
R86 0R
C71 47nF
C72 47nF
Q19
TBD
Q20
TBD
C78 4.7uF
C75 1000pF
R90 5mR
C86 1000pF
R94 5mR
MOTOR_A
STAT_LED
D3
STATUS LED
R75 330R
R68 22R
Y3
40Mhz
U12
ESP32-C3FH4
GPIO2/ADC1-2/FSPIQ
GPIO3/ADC1-3
GPIO8/STR
GPIO9/BOOT
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GPIO13/SPIWP
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GPIO15/SPICLK
GPIO16/SPID
GPIO17/SPIQ
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